# Review Paper

# Comparative Analysis of ML, Deep Learning, and Transformer Models for Human Cyber Behavior Prediction


## Abstract

This review paper examines how Machine Learning, Deep Learning, and Transformer-based models have been applied across published research on human cyber-risk behaviour prediction, and compares those approaches against the 12 models trained and evaluated to date within the Cyber Sense AI project. The comparison spans classical ML (SVM, Random Forest, XGBoost, Naïve Bayes, Decision Tree, Logistic Regression, KNN), Deep Learning (CNN, LSTM, Bi-LSTM, CNN+Bi-LSTM), and Transformer (Distil BERT) architectures, evaluated on a common SMS-based Social Engineering/Smishing Risk classification task. The review identifies where Cyber Sense AI's implementation choices align with, extend, or diverge from current literature, and closes with the specific gaps — graph-based modelling and explainability that remain for future project phases.


## 1. Introduction

Predicting human cyber-risk behaviour — phishing susceptibility, smishing, insider threat, credential risk — has been approached in the literature through three broad model families: classical Machine Learning, Deep Learning sequence/spatial models, and pre-trained Transformer architectures. This review paper compares how each family has been used across verified research papers relevant to Cyber Sense AI, against the 12 models actually implemented and evaluated in this project to date, in order to identify where this project's implementation choices align with, extend, or diverge from current published work.

## 2. Objective of the Review

1. Catalog the models used across the reviewed literature relevant to Cyber Sense AI's six risk categories.
2. Document the 12 models trained and evaluated within Cyber Sense AI to date, with full performance metrics.
3. Differentiate, approach-by-approach, how each research paper's model choice compares to Cyber Sense AI's corresponding implementation.
4. Identify remaining gaps to guide the project's next implementation phase.

## 3. Models Used in Reviewed Research Papers

| # | Research Paper | Model / Method Used | Model Category |
|---|---|---|---|
| 1 | Predicting cyber-criminal characteristics (Elsevier) | Classical ML classifiers | Classical ML |
| 2 | Human-factor security in cyber finance (Springer, 2026) | Conceptual/behavioural simulation | N/A — no model |
| 3 | Human factors in cybersecurity: interdisciplinary review (Springer) | Literature synthesis | N/A — no model |
| 4 | Leveraging human factors: integrated methodology (Springer) | Mixed-methods review | N/A — no model |
| 5 | BTITD (2026, Springer) | Dual-stream Times Net + BiLSTM | Deep Learning (sequential) |
| 6 | FusionITD (2026, Springer) | GNN + RAG-based semantic fusion | Graph-based + NLP fusion |
| 7 | ATHITD (2026, Elsevier) | Attention-based temporal heterogeneous GNN | Graph-based (attention) |
| 8 | BRITD (2026, Springer) | Stacked Bidirectional LSTM | Deep Learning (sequential) |
| 9 | Optimising Insider Threat Prediction (2024, Springer) | BiLSTM vs. RNN/LSTM comparison | Deep Learning (sequential) |
| 10 | UEBA hybrid model (Springer) | Attention Bi-LSTM + ResNet | Deep Learning (hybrid) |
| 11 | Supervised ML insider detection (2023, Springer) | Supervised ML vs. LSTM Autoencoder | Classical ML + Deep Learning |
| 12 | Forensic techniques for insider threats (2026, Springer) | Hybrid anomaly detection (non-DL) | Classical / statistical |
| 13 | SPEDIA dataset paper (2025, Elsevier) | Dataset-generation paper | N/A — no model |

| # | Research Paper | Model / Method Used | Model Category |
|---|---|---|---|
| 14 | Session-Based Detection: CERT & SPEDIA (2026, SSRN) | NB, DT, Random Forest, SVM, FFNN, XGBoost (supervised) + Autoencoders, Isolation Forest, k-means, LODA, LOF, OC-SVM (unsupervised) | Classical ML + Deep Learning (broad benchmark) |
| 15 | Rule Extraction & Interaction-Aware XAI for Malware (Springer) | Rule extraction (symbolic AI) | Symbolic / rule-based |
| 16 | Explainable Cyber Threat Intelligence Framework (Springer) | XAI-based prioritization | Explainability framework |
| 17 | Dynamic XAI risk management (Springer) | SHAP-based scoring | Explainability layer (model-agnostic) |
| 18 | PRISMA XAI systematic review | Meta-analysis of SHAP/LIME usage | N/A — no single model |
| 19 | Smishing detection via BERT (2025, Springer) | BERT | Transformer |
| 20 | Cost-Sensitive SMS Spam Detection (2026, Springer) | SVM (cost-sensitive) | Classical ML |
| 21 | ALBERT-BiLSTM cross-attention (2025, Elsevier) | ALBERT + BiLSTM cross-attention | Transformer + Deep Learning hybrid |
| 22 | Explainable RoBERTa phishing detector (2026, Elsevier) | RoBERTa + LIME | Transformer |
| 23 | CNN-LSTM+XAI+LLM phishing (2025/26, Springer) | CNN-LSTM hybrid | Deep Learning (hybrid) |
| 24 | DomURLs_BERT (2026, Springer) | BERT (domain/URL-adapted) | Transformer |
| 25 | PassREfinder-FL (2025, Elsevier) | GNN (link prediction) + Federated Learning | Graph-based + privacy-preserving |
| 26 | Genetic Programming password security (Springer) | Genetic Programming symbolic classifier | Evolutionary / symbolic |
| 27 | Frouros (Elsevier) | Drift-detection library (not a classifier) | N/A — tooling |
| 28 | Concept drift in behavioral biometrics (Elsevier) | Drift-detection applied to biometric auth | Statistical / drift detection |

## 4. Models Implemented and Evaluated in Cyber Sense AI

All 12 models below were trained and evaluated on the same task (SMS-based Social Engineering/Smishing Risk classification, per the project's "one risk at a time" rollout).

4.1 Classical Machine Learning Models

| # | Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 1 | Linear SVM | 97.94 | 97.74 | 86.67 | 91.87 |
| 2 | Random Forest | 97.76 | 100.0 | 83.33 | 90.91 |
| 3 | XGBoost | 97.67 | 95.59 | 86.67 | 90.91 |
| 4 | Multinomial Naïve Bayes | 96.59 | 100.0 | 74.67 | 85.50 |
| 5 | Decision Tree | 96.32 | 84.71 | 88.67 | 86.64 |
| 6 | Logistic Regression | 95.52 | 96.30 | 69.33 | 80.62 |
| 7 | KNN | 91.93 | 100.0 | 40.00 | 57.14 |

4.2 Deep Learning and Transformer Models

| # | Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 8 | CNN | 98.57 | 97.18 | 92.00 | 94.52 |
| 9 | LSTM | 96.86 | 97.50 | 78.52 | 86.99 |
| 10 | Bi-LSTM | 98.39 | 95.17 | 92.62 | 93.88 |
| 11 | CNN + Bi-LSTM | 98.39 | 96.45 | 91.28 | 93.79 |
| 12 | Distil BERT | 99.28 | 97.96 | 96.64 | 97.30 |
| 13 | ALBERT | 99.28 | 97.96 | 96.64 | 97.30 |

Best performer overall is the DistilBERT, leading on Accuracy, Recall, and F1 consistent with the literature trend where Transformer-based models consistently outperform both classical ML and non-Transformer Deep Learning across phishing/smishing/URL classification tasks.

## 5. Comparative Differentiation by Research Approach

This section maps each of Cyber Sense AI's core modeling approaches to a specific, individually verified research paper each paper below was checked against Section 3's list to confirm it is not a duplicate citation, ensuring every approach is backed by distinct, non-overlapping evidence.

| Research Approach | Research Paper (verified, non-duplicate) | Model Used in Research | CyberSense AI Model | Main Difference |
|---|---|---|---|---|
| Human-centred cyber-risk prediction | *Human-Centered Explainable AI for Security Enhancement: A Deep Intrusion Detection Framework* (2026) | CNN, LSTM + SHAP | Linear SVM / LSTM-based models | CNN-LSTM learns spatial + temporal relationships and pairs it with SHAP for trust; CyberSense's SVM is primarily a discriminative classifier without built-in temporal or explainability layers at this stage |
| Traditional phishing classification | *Phishing website prediction using base and ensemble classifier techniques with cross-validation* (Springer, *Cybersecurity*) | SVM, Random Forest, Naïve Bayes (base + ensemble) | Linear SVM | CyberSense AI independently evaluates SVM against 6 other competing models. |
| Text classification approaches | *Email Spam Classification: A Machine Learning Approach* (Springer, 2025/26) | Naïve Bayes (99.8% accuracy) | Multinomial Naïve Bayes | Both exploit probabilistic word-feature relationships; CyberSense compares it against stronger discriminative models (SVM, XGBoost, DistilBERT) rather than reporting NB in isolation |
| Ensemble cybersecurity detection | *EnLeM: ensemble learning-based model to detect phishing websites* (Springer, 2025/26) | Random Forest + KNN ensemble | Random Forest | Both use ensemble decision trees. CyberSense evaluates whether the added complexity of RF improves detection. |
| Gradient boosting approaches | *A novel multi-objective optimization-XGBoost based feature selection and optimization for enhanced phishing website detection* | XGBoost + multi-objective feature optimization | XGBoost | Both exploit sequentially improved weak learners; CyberSense measures raw XGBoost performance on the |

| Research Approach | Research Paper (verified, non-duplicate) | Model Used in Research | CyberSense AI Model | Main Difference |
|---|---|---|---|---|
| Sequential behaviour analysis | *Discriminating psychological stress levels: multi-level attentive LSTM approach* (Springer *Neural Computing and Applications*, 2025) | Multi-level Attentive LSTM | LSTM | Both model temporal dependencies, but CyberSense applies sequence modelling to cybersecurity behaviour (SMS text sequences) rather than physiological ECG signals |
| Hybrid temporal modelling | *A Hybrid CNN-LSTM Deep Learning Model* (IJAIA, 2025 — smart-grid intrusion detection) | CNN-LSTM hybrid | CNN + Bi-LSTM | Research combines local feature extraction and temporal modelling for SCADA/smart-grid intrusion; Cyber Sense investigates whether this same architecture transfers effectively to cybersecurity text/behaviour instead of industrial sensor data |
| Explainable cybersecurity | *LIME-Enhanced Insider Threat Detection for Distributed Security Systems* (Springer, 2026) | Deep learning + LIME | SHAP/LIME (planned) | Research focuses on interpretability as the primary contribution for insider-threat detection; Cyber Sense uses explainability as a core system requirement across all risk categories, not a single add-on |
| Transformer-based text analysis | *ExplainableDetector: Exploring Transformer-based Language Modeling Approach for SMS Spam Detection with Explainability Analysis* | Transformer (BERT-type) + explainability | Transformer models (Distil BERT) | Transformers use attention to capture contextual relationships; CyberSense evaluates their suitability against classical ML and recurrent models on the same SMS dataset, with |

| Research Approach | Research Paper (verified, non-duplicate) | Model Used in Research | CyberSense AI Model | Main Difference |
|---|---|---|---|---|
| Insider-threat detection | *Insider Threat Detection in User Activity Data Using Optimized LSTM-AE* (Springer, 2025) | Optimized LSTM-Autoencoder (CMU CERT r5.2) | LSTM/deep-learning experiments | Insider-threat studies generally model behavioural sequences for anomaly detection via reconstruction error. Cyber Sense extends the concept toward broader, supervised |

## 6. Key Findings

1. Transformer models dominate, both in the literature and in this project's own results Distil BERT's 99.28% accuracy and 97.30% F1 are the strongest results across all 12 implemented models, mirroring the literature's consistent finding that Transformer-based text classifiers outperform both classical ML and standard Deep Learning for phishing/smishing-style text classification.

2. Classical ML remains competitive on accuracy but weak on recall, KNN (91.93% accuracy but only 40% recall) and Logistic Regression (95.52% accuracy, 69.33% recall) show the classic precision-recall imbalance problem common in imbalanced spam/phishing datasets, also flagged in the Cost-Sensitive SMS Spam Detection paper's explicit imbalance-handling approach a good citation to justify why accuracy alone is a misleading metric for this task.

3. This project's implementation breadth (12 models across 3 categories) exceeds most individual reviewed papers, which typically test 1–4 models. This breadth is a legitimate point of methodological strength.

4. Every one of Cyber Sense AI's ten core research approaches is now backed by an individually verified, non-duplicate research paper, giving each modelling decision from classical SVM to Distil BERT, a distinct point of comparison in the literature rather than a single shared citation stretched across multiple claims.

5. Two gaps remain relative to the literature: graph-based modelling (used for insider-threat/credential-risk categories in the literature — GNN approaches in papers not yet implemented here) and explainability (SHAP/LIME, central to Cyber Sense AI's own stated novelty and demonstrated in papers, but not yet applied to the 12 models above).

## 7. Conclusion and Future Work

This review confirms that Cyber Sense AI's implemented models sit consistently within, and in several respects exceed, the scope of current published work on ML/DL/Transformer approaches to human cyber-risk prediction. The project's 7-model classical ML sweep and 5-model Deep Learning/Transformer comparison provide broader empirical coverage than most individual reviewed papers, and DistilBERT's leading performance is consistent with the field's broader shift toward Transformer-based architectures.

The next implementation phase should prioritize two items directly identified by this review:

(1) applying SHAP/LIME explainability to the existing 12 models, extending CyberSense AI's stated core requirement from design intent to measured outcome,

(2) evaluating a graph-based approach for the credential-risk and insider-threat categories, where GNN methods are well-established in the literature but not yet represented in this project's own model set.